End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B. Tech. CSE-AI, Semester: 6
(April -May, 2023)

Subject Code: BAI-322	Subject: Data Warehousing and Business Intelligence
Time :3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

(a) What do you mean by Metadata in Data warehouse?
(b) What are data marts?
(c) What are the essential differences between the MOLAP and ROLAP models?
(d) What is meant by slice and dice operation in OLAP?
(e) In data mining analysis, what are Type I and Type II errors?
(f) Differentiate between supervised learning and unsupervised learning? (With examples)
(g) Describe the Memory-based reasoning in Data mining?
(h) What are the different operations of Genetic Algorithms?

UNIT-I

Q2 How is a data warehouse different from the data base? How are they similar to each other? (10)

Q3 How are the top-down and bottom-up approaches for building a data warehouse different? Discuss the merits and demerits of each approach. (10)

UNIT-II

Q4 Explain the following OLAP operations with suitable example (10)
(a) Drill Down Analysis
(b) Slice and Dice
(c) Drill across
(d) Drill Through

Q5 What is a STAR schema? Discuss its component tables? (10)

UNIT-III

Q6 Discuss K-means Clustering Algorithm and Decision Tree Classifier in detail with a suitable example. (10)

Q7 What is meant by confusion matrix in machine learning and why is there a need for confusion matrix? Suppose we are trying to create a model that can predict the result for the cancer disease that is either a person has cancer or not. The confusion matrix is given as: (10)

n=100	Predicted: yes	Predicted: no
Actual: yes	28	7
Actual: no	4	~~77~~ 61

Explain the insights of the model in terms of Accuracy, Precision, Negative error rate, Sensitivity and Specificity.

UNIT-IV

Q8 Explain in detail the various applications of Data mining. (10)

Q9 Explain the association rule mining technique in detail. Also find the association rules for the following transactions X1, X2, X3 and X4. Given: Support= 60 % and Confidence =80%. (10)

T1: K,A,D,B
T2: D,A,C,E,B
T3: C,A,B,E
T4: B,A,D